



Panelization

V-groove • Mouse-bite tabs • Routed slots — combining boards for efficient assembly

A small DFM choice that saves 30-50% per board in production

Audience: Engineers preparing boards for production

Why panelize • Techniques • Tools • KiKit • Tab design • Edges



Why Panelize?

SMT lines process panels efficiently

Throughput

1 panel through SMT
= 10-30 boards.
Massively faster.

Cost

Per-board CM cost drops
30-50% by panelizing.

Fab pricing

Fab charges per panel area,
not per board.

Handling

Small boards (< 50 mm)
need panel to fit conveyor.

AOI / Test

Inspection equipment
works on panel size.

Fiducials

Shared panel fiducials
speed P&P alignment.

When to Panelize

Decision criteria

- ▶ Board $\leq 80 \times 50$ mm $\rightarrow$ panelize (otherwise inefficient on SMT line)
- ▶ Production quantity ≥ 100 pcs $\rightarrow$ almost always panelize
- ▶ Mass production (> 1000 pcs) $\rightarrow$ mandatory for cost
- ▶ Prototype quantity (< 50 pcs) $\rightarrow$ optional, may not be worth setup
- ▶ Board has fragile features (tabs, slots) $\rightarrow$ panelization adds support during assembly
- ▶ Multiple board variants $\rightarrow$ array different boards in one panel ('jewellery panel')
- ▶ Specific CM preference — ask which panel size they prefer (often 250×300 mm)
- ▶ Cost-sensitive product $\rightarrow$ panel cost savings = direct margin gain

Separation Techniques

Three ways to split boards from a panel

V-groove

Score top + bottom
Snap apart by hand or jig.
Fast, clean edge.

Mouse-bite tabs

Small drilled tabs
Break off → rough edge.
Flexible position.

Routed slots + tabs

CNC routs perimeter.
Leaves few break-tabs.
Cleanest edge.

Laser depanel

Laser-cuts perimeter.
No mechanical stress.
Premium method.

V-cut + tabs

Combine for very small boards.
V-cut between rows,
tabs between cols.

Pizza cutter

Older inline depanel.
Less common now.

V-Groove Depanelization

Score on top + bottom, snap on the line



- ▶ Score: 1/3 of board thickness on TOP, 1/3 on BOTTOM
- ▶ Result: 1/3 remains, easily snaps
- ▶ Score angle: 30° each side
- ▶ Component keep-out: ≥ 0.5 mm from groove line (silkscreen + parts)
- ▶ Best for: rectangular boards in straight rows/columns
- ▶ Limitation: only straight cuts — no curves
- ▶ Cost: minimal — just a saw blade on the panel
- ▶ Quality: clean edge, slight burrs (minor for most products)

Mouse-Bite Tabs

Drilled tab pattern — break apart

- ▶ Two parallel rows of small holes (0.5-1 mm dia) define the tab
- ▶ Tab area: typically 2-4 mm wide × board thickness
- ▶ Break by twisting board — tab fractures along the perforations
- ▶ Result: rough edge requiring sanding for cosmetic-critical products
- ▶ Best for: curved board outlines (where V-groove can't follow)
- ▶ Multiple tabs per side — typically 2-4 per long edge
- ▶ Avoid placing tabs near connectors / pads (they protrude on the broken edge)
- ▶ Cost: minimal — included in standard fab

Routed Slots + Break-Tabs

Premium method — CNC routes outline, leaves tabs

- ▶ CNC mill routs slots around each board, leaving 3-5 mm 'tabs'
- ▶ Tabs are 1-2 mm thick (vs 1.6 mm board) — easy to break
- ▶ Result: cleanest edge — basically pre-machined
- ▶ Best for: complex board outlines, premium products
- ▶ Allows curves, non-rectangular shapes
- ▶ Cost: +10-20% over V-groove (more machine time)
- ▶ Lead time: same as standard
- ▶ Combined with rail (panel perimeter) for handling support

Panel Rails + Fiducials

Common features around every panel

- ▶ Rails: 5-10 mm wide perimeter strip on top + bottom of panel
- ▶ Rails carry: tooling holes, fiducials, panel barcode, breakaway tabs
- ▶ Tooling holes: 3-4 mm dia, 4 per panel for assembly fixture alignment
- ▶ Fiducials × 3 minimum: 1 mm dia copper dots on opposite corners + center
- ▶ Panel barcode / part-number label printed on rail
- ▶ Rails get DISCARDED after assembly — don't put real circuitry there
- ▶ Rail width depends on CM equipment: 5 mm typical, 10 mm conservative
- ▶ KiKit / Altium / Cadence have automated rail generation

Tooling Holes

Mechanical alignment for the SMT line

- ▶ 4 tooling holes per panel — top-left, top-right, bottom-left, bottom-right corners
- ▶ Diameter: 3-4 mm (CM specifies)
- ▶ Plated or unplated — CM preference; unplated more common
- ▶ Position: usually on rails, NOT on individual boards
- ▶ Asymmetric placement: shift one hole inward to PREVENT panel flipping
- ▶ These holes mate to alignment pins on the SMT line conveyor
- ▶ Without tooling holes: panel CANNOT be processed by automated lines

KiKit — KiCad's Panelization Plugin

Free, open-source, capable

- ▶ Python script / KiCad plugin by Jan Mrázek (github.com/yaqwsx/KiKit)
- ▶ Supports: V-groove, mouse-bite tabs, routed slots, fiducials, rails
- ▶ CLI driven — easy to integrate into CI pipelines
- ▶ Reads a KiCad project + a YAML config → produces a panelized KiCad project
- ▶ Free, open-source — used by many KiCad professionals
- ▶ Alternatives: Altium ActiveBOM / Variants for panelization, Cadence Allegro
- ▶ Cost: 0 — versus commercial alternatives (KiCad has no built-in panelization)
- ▶ Learning curve: 1-2 days; many YouTube tutorials available

Common Panelization Mistakes

What CMs complain about

- ▶ V-groove on a board with curved outline → impossible, fab returns design
- ▶ Mouse-bite tabs too close to a connector → broken edge protrudes into pin area
- ▶ Insufficient component clearance from V-groove → component damaged on snap
- ▶ Missing tooling holes → CM can't process the panel — major delay
- ▶ Fiducials only on individual boards (none on panel) → vision system can't align
- ▶ Panel width > CM's max (often 350 mm) → won't fit through their oven
- ▶ Asymmetric panel can be flipped during loading — design with anti-rotation feature
- ▶ Designing the panel but submitting boards individually — wastes fab time

Cost Analysis Example

20 × 30 mm board, 1000 pcs production

- ▶ Single boards (no panelize): $1000 \times ₹25 \text{ fab} + 1000 \times ₹40 \text{ assembly} = ₹65\text{k total}$
- ▶ Panelized $4 \times 6 = 24$ boards per 200×200 mm panel:
- ▶ Need 42 panels (1008 boards) — $₹140 \text{ panel fab} + ₹400 \text{ panel assy}$
- ▶ $42 \times (₹140 + ₹400) = ₹22.7\text{k total}$
- ▶ Savings: ~65% per board
- ▶ Panel setup time on SMT line: ~10 min vs ~1 min per individual board
- ▶ Material waste: minimal — fab packs panels efficiently
- ▶ Trade-off: depanel labour (1-2 sec per board with V-groove jig)

Panelization Best Practices

Habits for successful production runs

- ▶ Talk to CM EARLY about preferred panel size + style
- ▶ Use V-groove for rectangular, mouse-bite for curves
- ▶ ≥ 0.5 mm component clearance from breakaway features
- ▶ Rails 5-10 mm wide with fiducials + tooling holes
- ▶ Panel symmetric WITH anti-rotation marker
- ▶ Use KiKit / commercial tool — don't manual-panelize
- ▶ Document panel separately (not just individual board files)
- ▶ Test panel separation on first sample boards
- ▶ Specify Class 2 / Class 3 to CM for tab quality
- ▶ Leave 3-5 mm between boards (allow router clearance)
- ▶ Plan for component overhang (cap to edge, etc.)
- ▶ Include panel barcode for traceability

Key Takeaways

Key takeaways from this lecture

Panelize ≥ 100 pcs

Saves 30-50% per board.

V-groove for rectangles

Cheapest, cleanest.

Mouse-bite for curves

Flexible position.

Rails + tooling holes

Required for SMT line.

Use KiKit (KiCad) / commercial tool

Avoid manual layout.

Talk to your CM

Their preferences shape the panel.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ KiKit — github.com/yaqwsx/KiKit
- ▶ JLCPCB / PCBWay panelization guides
- ▶ IPC-7711 — panel and depanel guidance
- ▶ Altium panelization docs
- ▶ Cadence Allegro panelization
- ▶ Phil's Lab YouTube — KiKit tutorial
- ▶ CM-specific design rules (Foxconn, Jabil, etc.)
- ▶ DesignSpark Mechanical — panel design

Thank you • Build something this week.